Evaluation runs: lb-ll (balancer-load-aware)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): l1: expected conforms (README); l2: expected conforms (README); l2-stale: expected violation (README says violation without a size rule. No verdict is given for k¿c && c¿=2 && m¿=2: a Reject there depends on k, q and m, and a model finds none at 3 1 2 2, 3 1 3 2, 4 1 2 3); l2-stale: expected conforms (at most k requests in flight, so no reject). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2,1,1,1	2	12	conforms	0.0	2	1 h, 250 GB	2	3	brain03, 2026-11-01
2,2,2	96	10240	conforms	0.8	2	1 h, 250 GB	96	4656	brain03, 2026-11-01
2,2,2,1	96	10240	conforms	0.8	2	1 h, 250 GB	96	4656	brain03, 2026-11-01
3,3,3			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01
3,3,3,2			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

stateless: pair **stale-poll**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2,1,1,1	6	148	32	violation	0.0	2	1 h, 250 GB	34	197	brain03, 2026-11-01
2,2,2,1	460			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

Not run (stateless). planned but never scheduled (grid points beyond the ladder): l1 2 2 3 fifo, l1 2 2 4 fifo, l1 2 2 5 fifo, l1 2 3 2 fifo, l1 2 3 3 fifo, l1 2 3 4 fifo, l1 2 3 5 fifo, l1 2 4 2 fifo, l1 2 4 3 fifo, l1 2 4 4 fifo, l1 2 4 5 fifo, l1 2 5 2 fifo, l1 2 5 3 fifo, l1 2 5 4 fifo, l1 2 5 5 fifo, l1 3 2 2 fifo, l1 3 2 3 fifo, l1 3 2 4 fifo, l1 3 2 5 fifo, l1 3 3 2 fifo, l1 3 3 4 fifo, l1 3 3 5 fifo, l1 3 4 2 fifo, l1 3 4 3 fifo, … (340 instances in all).

skipped: a smaller size had reached its limit: l1 4 4 4 fifo, l1 5 5 5 fifo, l1 6 6 6 fifo, l1 7 7 7 fifo, l1 8 8 8 fifo, l1 9 9 9 fifo, l2-stale 3 3 3 2 fifo, l2-stale 4 4 4 3 fifo, l2 4 4 4 3 fifo, l2-stale 5 5 5 4 fifo, l2 5 5 5 4 fifo, l2 6 6 6 5 fifo, l2-stale 6 6 6 5 fifo, l2 7 7 7 6 fifo, l2-stale 7 7 7 6 fifo, l2 8 8 8 7 fifo, l2-stale 8 8 8 7 fifo.

stateful

stateful: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,1,1,1	2	12	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-01
2,2,2	96	10240	conforms	0.6	2	1 h, 250 GB	brain02, 2026-11-01
2,2,2,1	96	10240	conforms	0.6	2	1 h, 250 GB	brain02, 2026-11-01
3,3,3			🕓 (48 h)			48 h, 250 GB	brain04, 2026-11-03
3,3,3,2			🕓 (1 h)			1 h, 250 GB	brain02, 2026-11-01

stateful: pair **stale-poll**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,1,1,1	6	148	32	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-01
2,2,2,1	460	7910400	2265344	violation	66.8	16	1 h, 250 GB	brain02, 2026-11-01
3,3,3,2				(1 h)			1 h, 250 GB	brain02, 2026-11-01

Not run (stateful). planned but never scheduled (grid points beyond the ladder): l1 2 2 3 fifo, l1 2 2 4 fifo, l1 2 2 5 fifo, l1 2 3 2 fifo, l1 2 3 3 fifo, l1 2 3 4 fifo, l1 2 3 5 fifo, l1 2 4 2 fifo, l1 2 4 3 fifo, l1 2 4 4 fifo, l1 2 4 5 fifo, l1 2 5 2 fifo, l1 2 5 3 fifo, l1 2 5 4 fifo, l1 2 5 5 fifo, l1 3 2 2 fifo, l1 3 2 3 fifo, l1 3 2 4 fifo, l1 3 2 5 fifo, l1 3 3 2 fifo, l1 3 3 4 fifo, l1 3 3 5 fifo, l1 3 4 2 fifo, l1 3 4 3 fifo, … (340 instances in all).

skipped: a smaller size had reached its limit: l1 4 4 4 fifo, l1 5 5 5 fifo, l1 6 6 6 fifo, l1 7 7 7 fifo, l1 8 8 8 fifo, l1 9 9 9 fifo, l2-stale 4 4 4 3 fifo, l2 4 4 4 3 fifo, l2-stale 5 5 5 4 fifo, l2 5 5 5 4 fifo, l2 6 6 6 5 fifo, l2-stale 6 6 6 5 fifo, l2 7 7 7 6 fifo, l2-stale 7 7 7 6 fifo, l2 8 8 8 7 fifo, l2-stale 8 8 8 7 fifo.

offline

offline: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2,2	96	10240	conforms	0.7	8	10 min, 16 GB	0.1	0.7	0.0	in-memory	brain05, 2026-11-02

offline: pair **correct** against **l2**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	2	12	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2,1	96	10240	conforms	0.7	8	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain05, 2026-11-02

offline: pair **correct** against **l2-stale**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	6	12	conforms	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2,1	460	10240	conforms	0.6	8	10 min, 16 GB	0.1	0.6	0.0	in-memory	brain05, 2026-11-02

offline: pair **stale-poll** against **l2**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	2	148	2	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2,1	96	7910400	284	violation	83.9	22	10 min, 16 GB	0.1	83.9	0.0	in-memory	brain05, 2026-11-02

offline: pair **stale-poll** against **l2-stale**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	6	148	2	violation	0.1	8	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2,2,2,1	460	7910400	284	violation	83.9	22	10 min, 16 GB	0.1	83.9	0.0	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): l1 2 2 3 fifo, l1 2 2 4 fifo, l1 2 2 5 fifo, l1 2 3 2 fifo, l1 2 3 3 fifo, l1 2 3 4 fifo, l1 2 3 5 fifo, l1 2 4 2 fifo, l1 2 4 3 fifo, l1 2 4 4 fifo, l1 2 4 5 fifo, l1 2 5 2 fifo, l1 2 5 3 fifo, l1 2 5 4 fifo, l1 2 5 5 fifo, l1 3 2 2 fifo, l1 3 2 3 fifo, l1 3 2 4 fifo, l1 3 2 5 fifo, l1 3 3 2 fifo, l1 3 3 4 fifo, l1 3 3 5 fifo, l1 3 4 2 fifo, l1 3 4 3 fifo, … (340 instances in all).

skipped: a smaller size had reached its limit: l1 3 3 3 fifo, l1 4 4 4 fifo, l1 5 5 5 fifo, l1 6 6 6 fifo, l1 7 7 7 fifo, l1 8 8 8 fifo, l1 9 9 9 fifo, l2 3 3 3 2 fifo, l2-stale 3 3 3 2 fifo, l2 4 4 4 3 fifo, l2-stale 4 4 4 3 fifo, l2-stale 5 5 5 4 fifo, l2 5 5 5 4 fifo, l2 6 6 6 5 fifo, l2-stale 6 6 6 5 fifo, l2-stale 7 7 7 6 fifo, l2 7 7 7 6 fifo, l2 8 8 8 7 fifo, l2-stale 8 8 8 7 fifo.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2,2	96	10240	conforms	57.6	1176	10 min, 16 GB	0.1	42.9	14.7	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **correct** against **l2**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	2	12	conforms	3.2	172	10 min, 16 GB	0.1	3.2	0.0	in-memory	brain01, 2026-11-02
2,2,2,1	96	10240	conforms	44.2	977	10 min, 16 GB	0.1	44.2	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **correct** against **l2-stale**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	6	12	conforms	3.2	172	10 min, 16 GB	0.1	3.2	0.0	in-memory	brain01, 2026-11-02
2,2,2,1	460	10240	conforms	44.2	977	10 min, 16 GB	0.1	44.2	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **stale-poll** against **l2**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	2	148	2	violation	3.6	160	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **stale-poll** against **l2-stale**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	6	148	2	violation	3.6	160	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): l1 2 2 3 fifo, l1 2 3 2 fifo, l1 3 2 2 fifo, l2-stale 2 1 1 2 fifo, l2-stale 2 1 2 1 fifo, l2-stale 2 2 1 1 fifo, l2 2 2 2 2 fifo, l2 2 2 3 1 fifo, l2 2 3 2 1 fifo, l2-stale 3 1 1 1 fifo, l2 3 2 2 1 fifo.

skipped: a smaller size had reached its limit: l1 3 3 3 fifo, l2-stale 2 2 2 1 fifo.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	2	12	conforms	3.0	168	10 min, 16 GB	3.0	0.1	0.0	in-memory	brain01, 2026-11-01
2,2,2	96	10240	conforms	3.6	164	10 min, 16 GB	3.6	0.7	0.0	in-memory	brain01, 2026-11-01
2,2,2,1	96	10240	conforms	3.6	171	10 min, 16 GB	3.6	0.6	0.0	in-memory	brain01, 2026-11-01
2,3,2	1280	925696	conforms	54.4	1986	48 h, 80 GB	20.0	54.2	0.1	merge	pinky21, 2026-11-07

cross offline, Rust implementation against the Java specification: pair **stale-poll**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1,1,1	2	148	2	violation	3.0	168	10 min, 16 GB	3.0	0.1	0.0	in-memory	brain01, 2026-11-01
2,2,2,1	96	7910400	284	violation	83.9	185	10 min, 16 GB	3.6	83.9	0.0	in-memory	brain01, 2026-11-01

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): l1 2 2 3 fifo, l1 3 2 2 fifo, l2-stale 2 2 2 2 fifo, l2 2 2 2 2 fifo, l2 2 2 3 1 fifo, l2-stale 2 2 3 1 fifo, l2 2 3 2 1 fifo, l2-stale 2 3 2 1 fifo, l2-stale 3 2 2 1 fifo, l2 3 2 2 1 fifo.

left out by the table rules (ok): l2-stale 2 1 1 1 fifo, l2 2 1 1 1 fifo, l2-stale 2 2 2 1 fifo, l2 2 2 2 1 fifo.

skipped: a smaller size had reached its limit: l1 3 3 3 fifo.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,2,2	96	10240	conforms	82.0	343	1 h, 250 GB		brain05, 2026-11-02
3,3,3	33067440		(1 h)			1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **correct** against **l2**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1,1,1	2	12	conforms	0.8	152	1 h, 250 GB		brain05, 2026-11-02
2,2,2,1	96	10240	conforms	68.2	358	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **correct** against **l2-stale**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1,1,1	6	12	conforms	0.7	146	1 h, 250 GB		brain05, 2026-11-02
2,2,2,1	460	10240	conforms	66.9	356	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **stale-poll** against **l2**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1,1,1	2	148	32	violation	1.4	171	1 h, 250 GB	170	brain05, 2026-11-02
2,2,2,1	96			(1 h)			1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **stale-poll** against **l2-stale**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1,1,1	6	148	32	violation	1.2	172	1 h, 250 GB	145	brain05, 2026-11-02
2,2,2,1	460			(1 h)			1 h, 250 GB		brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): l1 2 2 2 fifo, l1 3 3 3 fifo, l2-stale 2 1 1 1 fifo, l2 2 1 1 1 fifo, l2 2 2 2 1 fifo, l2-stale 2 2 2 1 fifo.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1,1,1	2	12	conforms	3.0	164	1 h, 250 GB		brain04, 2026-11-01
2,2,2	96	10240	conforms	4.4	160	1 h, 250 GB		brain04, 2026-11-01
2,2,2,1	96	10240	conforms	4.4	167	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **stale-poll**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1,1,1	2	148	32	violation	3.0	164	1 h, 250 GB	2	brain04, 2026-11-01
2,2,2,1	96	7910400	2265344	violation	86.8	167	1 h, 250 GB	3	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): l1 2 2 2 fifo, l2-stale 2 1 1 1 fifo, l2 2 1 1 1 fifo, l2 2 2 2 1 fifo, l2-stale 2 2 2 1 fifo.

skipped: a smaller size had reached its limit: l1 3 3 3 fifo.